

LAMIA, Greece

WMO: 166750

Lat: 38.8767N

Lon: 22.4361E

Elev: 41

StdP: 14.67

Time Zone: 2.00 (EUE)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.5	31.7	18.5	14.0	37.7	22.4	16.9	38.3	23.5	52.9	19.9	46.8	6.0	290	0.423

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.6	96.0	69.6	93.6	69.4	91.5	68.8	72.5	88.4	71.5	87.9	70.5	86.5	9.0	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
68.2	103.9	76.2	66.9	99.0	75.9	65.6	94.6	75.6	36.1	87.4	35.2	88.1	34.4	86.1	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.5	17.3	14.5		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.7	45.8	46.8	50.9	58.3	67.0	75.9	78.2	78.2	72.4	63.4	54.9	48.1
	DBStd	13.10	5.54	6.17	5.88	5.26	5.45	4.77	3.89	3.32	5.01	5.95	6.23	5.50
	HDD50	463	152	120	63	4	0	0	0	0	0	0	27	97
	HDD65	2735	596	509	439	209	43	1	0	0	7	101	307	523
	CDD50	4746	20	32	90	253	528	776	873	873	673	415	173	40
	CDD65	1541	0	0	0	8	106	327	408	408	230	51	3	0
	CDH74	16328	0	0	1	114	1074	3659	4598	4311	2160	393	18	0
	CDH80	7438	0	0	0	19	305	1673	2341	2164	853	81	2	0
(14) Wind	WSAvg	5.9	5.3	5.9	6.0	6.5	6.7	7.0	6.3	6.1	6.0	4.9	4.6	5.1
(15) Precipitation	PrecAvg	20.6	2.3	2.0	2.1	1.4	1.5	0.6	0.7	0.7	1.3	2.6	2.9	2.6
	PrecMax	29.0	5.8	4.1	5.0	3.7	2.8	2.9	2.6	3.3	3.6	9.5	10.1	6.0
	PrecMin	13.9	0.2	0.2	0.7	0.1	0.2	0.0	0.0	0.0	0.0	0.1	0.4	0.0
	PrecStd	3.4	1.5	1.1	1.1	0.9	0.8	0.6	0.7	0.9	0.9	1.9	2.1	1.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.0	68.8	73.0	83.1	90.4	98.6	99.8	97.7	94.8	86.6	75.8	68.4
		MCWB	52.7	55.3	56.0	61.1	66.5	69.5	71.3	69.9	68.2	65.4	59.7	57.3
	2%	DB	60.7	64.7	69.4	77.4	86.2	94.7	96.1	95.0	90.8	82.2	71.3	63.9
		MCWB	51.9	53.5	55.2	59.4	64.7	68.7	70.0	70.0	67.8	64.7	59.1	54.6
	5%	DB	57.9	61.7	66.6	73.3	83.2	91.8	93.5	92.7	88.0	78.3	68.2	59.8
		MCWB	50.7	52.1	54.0	58.2	64.1	68.4	70.0	69.3	66.8	63.5	58.0	52.1
	10%	DB	55.4	58.4	63.2	70.2	80.2	89.1	91.0	90.6	85.4	75.2	64.8	57.0
		MCWB	49.8	50.3	52.5	56.9	63.1	67.7	69.2	68.9	66.2	62.5	57.2	50.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	55.5	57.1	58.9	62.6	69.2	72.8	73.7	73.8	72.2	68.7	63.3	58.7
		MCDB	60.3	65.7	68.6	76.6	84.2	89.9	90.0	87.7	84.4	79.8	69.2	64.8
	2%	WB	53.8	54.9	56.6	61.1	67.0	71.1	72.4	72.6	70.1	66.7	61.1	55.4
		MCDB	58.6	62.4	65.9	74.2	81.8	89.6	89.8	87.3	83.9	76.6	67.5	61.8
	5%	WB	51.9	53.2	55.1	59.6	65.4	69.9	71.4	71.5	68.8	65.3	59.6	53.5
		MCDB	56.2	60.3	64.6	71.0	79.2	87.4	89.0	86.4	82.7	74.5	65.9	58.3
	10%	WB	50.3	51.1	53.3	58.0	64.1	68.8	70.4	70.5	67.7	63.9	58.1	51.6
		MCDB	54.5	57.2	61.9	67.8	76.9	85.7	87.0	85.3	81.3	72.5	64.0	56.0
(35) Mean Daily Temperature Range	5% DB	MDBR	15.5	17.1	17.6	20.4	21.0	23.7	24.6	24.3	23.4	18.6	17.3	15.5
		MCDBR	20.9	23.9	25.0	26.0	26.4	27.6	28.6	27.3	28.0	25.1	22.3	20.7
		MCWBR	13.2	14.0	13.2	11.8	10.4	9.2	8.8	9.1	9.5	10.7	11.6	12.7
	5% WB	MCDBR	16.2	21.0	22.3	24.5	24.2	24.7	25.6	23.5	24.8	21.1	19.8	17.7
		MCWBR	10.8	13.0	12.8	12.0	10.5	9.3	8.6	9.0	9.9	9.9	10.9	11.3
(40) Clear-Sky Solar Irradiance	taub		0.377	0.412	0.464	0.518	0.503	0.483	0.491	0.502	0.480	0.445	0.410	0.384
	taud		2.308	2.168	2.026	1.910	2.011	2.116	2.103	2.065	2.102	2.166	2.221	2.266
	Ebn at Noon		246	251	249	244	251	256	252	246	242	237	230	231
	Edh at Noon		30	39	50	60	55	49	50	51	46	39	33	30
(44) All-Sky Solar Radiation	RadAvg		611	850	1236	1635	1951	2238	2321	2078	1528	1035	710	538
	RadStd		88	98	114	152	105	146	106	86	112	108	72	83

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (12 neighbors)		+0.63	N/A	N/A	+1.12	N/A	N/A	N/A	-149	+175	N/A	

Nomenclature: See separate page